

Mathematics I

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Numerically small Entities

Consider a geometric square of side length a_0 . The area A is determined as

$$A = a_0^2$$

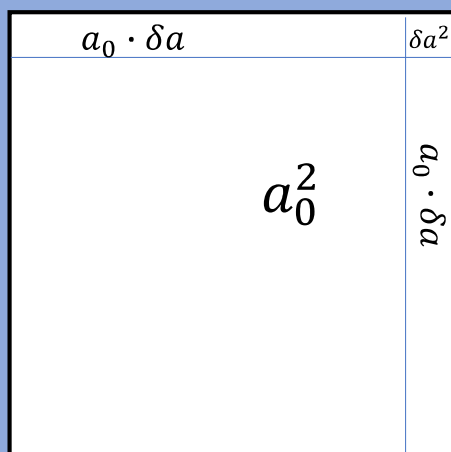
If the side length of the square is changed slightly to $a = a_0 + \delta a$ (note that δa may be positive or negative), then the expression of the area changes to

$$A = (a_0 + \delta a)^2 = a_0^2 + 2a_0 \cdot \delta a + \delta a^2$$

It is evident, that the area is changed by the change of side length.

This change may be presented in graphical form as follows:

$$A = (a_0 + \delta a)^2 = a_0^2 + 2a_0 \cdot \delta a + \delta a^2$$



contribution of 1st order



contribution
of 2nd order



contribution of 1st order

If the change of side length is small then there will also be a relatively small change of the area. Up to first order (that is those contributions which are proportional to δa^0 (0th order) or δa^1 (1st order)) have:

$$A \approx a_0^2 + 2a_0 \cdot \delta a + \text{ord}(\delta a^2)$$

here $\text{ord}(\delta a^2)$ includes contributions of higher order, namely those proportional to δa^2 or higher.

The change in area, δA , is, to this approximation

$$A + \delta A \approx a_0^2 + 2a_0 \cdot \delta a \quad \Rightarrow \quad \delta A \approx 2a_0 \cdot \delta a$$

and therefore $\frac{\delta A}{\delta a} \approx 2a_0$

Consider the limit of δa becoming very small such, that it tends towards 0. This is expressed as:

$$\lim_{\delta a \rightarrow 0} \frac{\delta A}{\delta a} = 2a_0$$

Note that all contributions of higher order in δa tend towards 0 in this limit.

This process of taking the limit results in the **derivative**. It may be obtained using a limiting process by considering a **difference quotient**.

$$\lim_{\delta x \rightarrow 0} \frac{y(x) - y(x + \delta x)}{x - (x + \delta x)} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{d}{dx} y(x) = y'(x)$$

Calculation for $y(x) = x^2$

$$\lim_{\delta x \rightarrow 0} \frac{y(x) - y(x + \delta x)}{x - (x + \delta x)} = \lim_{\delta x \rightarrow 0} \frac{x^2 - (x + \delta x)^2}{-\delta x} =$$

$$\lim_{\delta x \rightarrow 0} \frac{x^2 - (x^2 + 2x\delta x + \delta x^2)}{-\delta x} = \lim_{\delta x \rightarrow 0} \frac{x^2 - x^2 - 2x\delta x - \delta x^2}{-\delta x} =$$

$$\lim_{\delta x \rightarrow 0} \frac{2x\delta x + \delta x^2}{\delta x} = \lim_{\delta x \rightarrow 0} (2x + \delta x) = 2x$$

Therefore have:

$$\frac{d}{dx} y(x) = \frac{d}{dx} x^2 = 2x$$

Calculation for $y(x) = x^3$

$$\lim_{\delta x \rightarrow 0} \frac{y(x + \delta x) - y(x)}{(x + \delta x) - x} = \lim_{\delta x \rightarrow 0} \frac{(x + \delta x)^3 - x^3}{\delta x} =$$

$$\lim_{\delta x \rightarrow 0} \frac{(x^3 + 3x^2\delta x + 3x^1\delta x^2 + \delta x^3) - x^3}{\delta x} = \lim_{\delta x \rightarrow 0} (3x^2 + 3x^1\delta x + \delta x^2)$$

$$= 3x^2$$

Therefore have:

$$\frac{d}{dx}y(x) = \frac{d}{dx}x^3 = 3x^2$$

Calculation for $y(x) = x^n$

$$\lim_{\delta x \rightarrow 0} \frac{y(x + \delta x) - y(x)}{(x + \delta x) - x} = \lim_{\delta x \rightarrow 0} \frac{(x + \delta x)^n - x^n}{\delta x} =$$

$$\lim_{\delta x \rightarrow 0} \frac{(x^n + nx^{n-1}\delta x + ord(\delta x^2)) - x^n}{\delta x} = \lim_{\delta x \rightarrow 0} (nx^{n-1} + ord(\delta x))$$

$$= nx^{n-1}$$

Therefore have:

$$\frac{d}{dx} y(x) = \frac{d}{dx} x^n = nx^{n-1}$$

Calculation for $y(x) = \sin(x)$

$$\lim_{\delta x \rightarrow 0} \frac{y(x + \delta x) - y(x)}{(x + \delta x) - x} = \lim_{\delta x \rightarrow 0} \frac{\sin(x + \delta x) - \sin(x)}{\delta x}$$

Use the following identity:

$$\begin{aligned} \sin(\alpha + \beta) &= \sin(\alpha) \cdot \cos(\beta) + \cos(\alpha) \cdot \sin(\beta) \\ \lim_{\delta x \rightarrow 0} \frac{\sin(x) \cdot \cos(\delta x) + \cos(x) \cdot \sin(\delta x) - \sin(x)}{\delta x} \\ &= \sin(x) \cdot \lim_{\delta x \rightarrow 0} \frac{\cos(\delta x) - 1}{\delta x} + \cos(x) \cdot \lim_{\delta x \rightarrow 0} \frac{\sin(\delta x)}{\delta x} \end{aligned}$$

It is possible to show that

$$\lim_{\delta x \rightarrow 0} \frac{(\cos(\delta x) - 1)}{\delta x} = 0 \text{ und } \lim_{\delta x \rightarrow 0} \frac{\sin(\delta x)}{\delta x} = 1$$

Therefore have:

$$\frac{d}{dx} \sin(x) = \frac{d}{dx} \sin(x) = \cos(x)$$

Differentiation, integration and the antiderivative

It is possible to reverse the process of differentiation. The reverse process is called **integration**, or occasionally also **antidifferentiation**.

Example: when $y = 3x + 5$, then $\frac{d}{dx}y = \frac{d}{dx}(3x + 5) = 3$

The reverse process: $\int 3 dx = 3x + C$ here C is a constant (the constant of integration).

The constant of integration is fixed by the **boundary conditions**. For example the integrated function has to have the value of 20 for $x=5$. This fixes the value of C as: $C=5$ as $3*5+5=20$.

Differentiation of Sums of Functions

Let $f(x)$ and $g(x)$ be functions, and a and b constants.

Also: $a, b \in \mathbb{R}$. Then have:

$$\frac{d}{dx}(a \cdot f(x)) = a \cdot \frac{d}{dx}f(x)$$

$$\frac{d}{dx}(a \cdot f(x) + b \cdot g(x)) = a \cdot \frac{d}{dx}f(x) + b \cdot \frac{d}{dx}g(x)$$

$$\frac{d}{dx}(a \cdot f(x) - b \cdot g(x)) = a \cdot \frac{d}{dx}f(x) - b \cdot \frac{d}{dx}g(x)$$

Differentiation of the Reciprocal of a Function

Let $f(x)$ be a function. Then have:

$$\begin{aligned}\frac{d}{dx}\left(\frac{1}{f(x)}\right) &= \lim_{\delta x \rightarrow 0} \frac{\frac{1}{f(x + \delta x)} - \frac{1}{f(x)}}{\delta x} \\&= \lim_{\delta x \rightarrow 0} \left[\frac{1}{f(x + \delta x)} \cdot \frac{1}{f(x)} \cdot \left(-\frac{f(x + \delta x) - f(x)}{\delta x} \right) \right] \\&= -\frac{\frac{d}{dx}f(x)}{(f(x))^2}\end{aligned}$$

Quotient Rule

Differentiation of Products of Functions

Let $f(x)$ and $g(x)$ be functions. Then have:

$$\begin{aligned}\frac{d}{dx}(f(x) \cdot g(x)) &= \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) \cdot g(x + \delta x) - f(x) \cdot g(x)}{(x + \delta x) - x} \\&= \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) \cdot g(x + \delta x) - f(x) \cdot g(x + \delta x) + f(x) \cdot g(x + \delta x) - f(x) \cdot g(x)}{\delta x} \\&= \lim_{\delta x \rightarrow 0} \left(\frac{f(x + \delta x) - f(x)}{\delta x} \right) \cdot g(x + \delta x) + f(x) \cdot \lim_{\delta x \rightarrow 0} \left(\frac{g(x + \delta x) - g(x)}{\delta x} \right) = \\&= \left(\frac{d}{dx} f(x) \right) \cdot g(x) + f(x) \cdot \left(\frac{d}{dx} g(x) \right)\end{aligned}$$

Product Rule

Differentiation of Ratios of Functions

Let $f(x)$ and $g(x)$ be functions. Then have:

$$\begin{aligned}\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) &= \left(\frac{d}{dx} f(x) \right) \cdot \frac{1}{g(x)} + f(x) \left(\frac{d}{dx} \frac{1}{g(x)} \right) = \\ &= \left(\frac{d}{dx} f(x) \right) \cdot \frac{1}{g(x)} - \frac{f(x)}{(g(x))^2} \cdot \left(\frac{d}{dx} g(x) \right) = \\ &= \frac{f' \cdot g - g' \cdot f}{g^2}\end{aligned}$$

Rule for the differentiation of $\frac{f(x)}{g(x)}$

Differentiation of Functions in Functions

Let $f(x)$ and $g(x)$ be functions. Then have:

$$\frac{d}{dx}(f(g(x))) = \lim_{\delta x \rightarrow 0} \frac{f(g(x + \delta x)) - f(g(x))}{\delta x}$$

Multiply with 1

$$= \lim_{\delta x \rightarrow 0} \frac{f(g(x + \delta x)) - f(g(x))}{g(x + \delta x) - g(x)} \cdot \frac{g(x + \delta x) - g(x)}{\delta x}$$
$$= \left(\frac{d}{dy} f(y) \right) \cdot \left(\frac{d}{dx} g(x) \right)$$

Chain Rule

Examples

$$\frac{d}{dx}(2 \cdot x^3 - 5 \cdot x) = 6x^2 - 5$$

$$\frac{d}{dx}(2 \cdot x^8) = 16 \cdot x^7$$

or alternatively

$$\begin{aligned} \frac{d}{dx}(2 \cdot x^3 \cdot x^5) &= 2 \cdot \left(\frac{d}{dx}x^3\right) \cdot x^5 + 2 \cdot x^3 \cdot \left(\frac{d}{dx}x^5\right) = 2 \cdot 3 \cdot x^2 \cdot x^5 + 2 \cdot x^3 \cdot 5 \cdot x^4 = (6 + 10) \cdot x^7 = 16 \cdot x^7 \end{aligned}$$

$$\begin{aligned} \frac{d}{dx}\left(\frac{x^5 + 4}{x^3 - 3}\right) &= \frac{5 \cdot x^4 \cdot (x^3 - 3) - (x^5 + 4) \cdot 3 \cdot x^2}{(x^3 - 3)^2} \\ &= \frac{5 \cdot x^7 - 15 \cdot x^4 - 3 \cdot x^7 - 12 \cdot x^2}{(x^3 - 3)^2} = \frac{2 \cdot x^7 - 15 \cdot x^4 - 12 \cdot x^2}{(x^3 - 3)^2} \end{aligned}$$

Examples

$$\begin{aligned} \frac{d}{dx} \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \dots \right) \\ = 0 + 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \dots \end{aligned}$$

There is one other function, for which the derivative is equal to the function itself as for:

$$\frac{d}{dx} e^x = e^x$$

Here $e = 2.71828$. It is possible to show that:

$$e^x = \sum_{j=0}^{\infty} \frac{x^j}{j!} = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \dots$$

Differentiation of the Inverse Function

Let $y = f(x)$ and $f^{-1}(y) = x$. Then have: $f^{-1}(f(x)) = x$

Calculate derivative:

$$\frac{d}{dx} f^{-1}(f(x)) = \frac{d}{dx} x = 1$$

$$\left(\frac{d}{dy} f^{-1}(y) \right) \cdot \left(\frac{d}{dx} f(x) \right) = 1$$

Example 1

Let $f(x) = e^x$ and the inverse function f^{-1} is $\ln(x)$.

Obviously $e^{\ln(x)} = x$ und $\ln(e^x) = x$.

Calculate derivative. Let $x = \ln(y)$ and $y = e^x$

$$\frac{d}{dx} y = \frac{d}{dx} f(x) = \frac{d}{dx} e^x = e^x = y$$

$$\text{As } \left(\frac{d}{dy} f^{-1}(y) \right) \cdot \left(\frac{d}{dx} f(x) \right) = 1$$

one must conclude that: $\frac{d}{dy} f^{-1}(y) = \frac{1}{y}$

$$\text{Therefore } \frac{d}{dx} \ln(x) = \frac{1}{x}$$

Example 2

Let $f(x) = \sin(x)$ and the inverse function f^{-1} is $\arcsin(x)$.
Obviously $\sin(\arcsin(x)) = x$ und $\arcsin(\sin(x)) = x$.

Calculate derivative. Let $y = \sin(x)$ and $x = \arcsin(y)$

$$\frac{d}{dx} y = \cos(x) = \sqrt{1 - \sin(x)^2} = \sqrt{1 - y^2}$$

$$\frac{d}{dy} x = \frac{d}{dy} \arcsin(y) = \frac{1}{\frac{d}{dx} y} = \frac{1}{\sqrt{1 - y^2}}$$

Then $\left(\frac{d}{dy} f^{-1}(y) \right) \cdot \left(\frac{d}{dx} f(x) \right) = 1$ demands that

$$\frac{d}{dx} \arcsin(x) = \frac{1}{\sqrt{1-x^2}} \quad (\text{in standard notation})$$

